



CSE-476: Introduction to Natural Language Processing

Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In the BIO tagging scheme for NER, what does the "B-" prefix indicate?
- A. Beginning of an entity
 - B. Inside an entity
 - C. Outside any entity
 - D. Between two tokens
- Q2. Which statement best describes a (plain) Markov chain in sequence tagging?
- A. It uses only words (observations) and ignores labels
 - B. It is identical to an HMM
 - C. It models transitions between labels but does not consider the words
 - D. It models emissions $P(\text{word} \mid \text{label})$
- Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:
- A. $P(\text{label} \mid \text{previous label})$
 - B. $P(\text{label} \mid \text{word})$
 - C. $P(\text{word} \mid \text{previous word})$
 - D. $P(\text{word} \mid \text{label})$
- Q4. Which of the following best describes Semantic Role Labeling (SRL)?:
- A. A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location
 - B. A process that assigns grammatical roles like subject and object based on syntactic structure

- C. A method that classifies named entities and links them to a knowledge base
 - D. A technique that labels words with their parts of speech and morphological features
- Q5. Which of the following best describes Byte Pair Encoding (BPE)?
- A. A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
 - B. A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
 - C. A tokenization method that splits text based on whitespace and punctuation
 - D. A character-level encoding scheme that assigns fixed-length binary codes to characters
- Q6. Which of the following best describes the main goal of tokenization in NLP?
- A. To label each token with a grammatical category
 - B. To convert text into a sequence of symbols suitable for modeling
 - C. To reduce vocabulary size to zero
 - D. To assign probabilities to sentences
- Q7. Why is splitting text by spaces (`text.split(" ")`) considered an insufficient tokenizer?
- A. It produces too many tokens
 - B. It does not work well for languages without explicit word boundaries
 - C. It cannot tokenize English at all
 - D. It ignores punctuation entirely
- Q8. Which of the following is NOT listed as a common sequence tagging task?
- A. Part-of-speech tagging
 - B. Named Entity Recognition
 - C. Dependency parsing
 - D. Sentiment analysis
- Q9. What is a common limitation of BIO tagging for Named Entity Recognition?
- A. It cannot represent overlapping or nested entities
 - B. It requires too many labels
 - C. It does not support multiple entity types
 - D. It ignores entity boundaries

- Q10. When choosing between character-level tokenization and sentence-level tokenization, which trade-off must be considered?
- A. Speed vs accuracy
 - B. Tokenized sequence length vs vocabulary size
 - C. Precision vs recall
 - D. Memory vs computation